

Applications of Surface Integrals; Flux Worksheet

1. Let σ be the portion of the cylinder $x^2 + y^2 = 16$ between the planes $z = -2$ and $z = 2$ oriented by outward normal. Compute the flux of $\mathbf{F}$ across σ , where $\mathbf{F}(x, y, z) = \langle 3, -7, z \rangle$.

We see that we cannot write this surface as a function $z = f(x, y)$ and therefore we must approach this question by viewing the surface parametrically.

We create a parameterization for σ :

$$\mathbf{r}(u, v) = \langle 4 \cos(u), 4 \sin(u), v \rangle \text{ where } 0 \leq u \leq 2\pi \text{ \& } -2 \leq v \leq 2.$$

We calculate that:

$$\mathbf{r}_u = \langle -4 \sin(u), 4 \cos(u), 0 \rangle$$

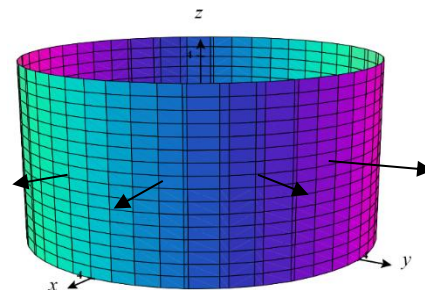
$$\mathbf{r}_v = \langle 0, 0, 1 \rangle$$

Therefore,

$$\begin{aligned} \mathbf{r}_u \times \mathbf{r}_v &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 \sin(u) & 4 \cos(u) & 0 \\ 0 & 0 & 1 \end{vmatrix} \\ &= \langle 4 \cos(u), 4 \sin(u), 0 \rangle \end{aligned}$$

Now, we have that,

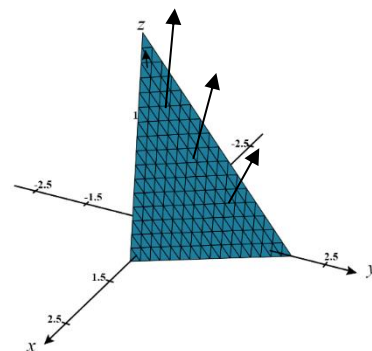
$$\begin{aligned} \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA \\ &= \int_{-2}^2 \int_0^{2\pi} \langle 3, -7, v \rangle \cdot \langle 4 \cos(u), 4 \sin(u), 0 \rangle \, du \, dv \\ &= \int_{-2}^2 \int_0^{2\pi} [12 \cos(u) - 28 \sin(u)] \, du \, dv \\ &= \int_{-2}^2 \left[12 \sin(u) + 28 \cos(u) \right]_0^{2\pi} \, dv \\ &= \int_{-2}^2 [(0 + 28) - (0 + 28)] \, dv \\ &= 0 \end{aligned}$$



2. Let σ be the portion of the plane $6x + 3y + 2z = 6$ in the first octant oriented by upwards normal and $\mathbf{F}(x, y, z) = \langle x^2, yx, zx \rangle$. Compute the flux of $\mathbf{F}$ across σ in the positive direction.

Since σ represents the graph of a function of the form $z = f(x, y)$, specifically $z = 3 - 3x - \frac{3}{2}y$, we do not have to find a parameterization for σ and instead we tackle the problem as follows.

We note the function $G(x, y, z) = z - \left(3 - 3x - \frac{3}{2}y\right) = z - 3 + 3x + \frac{3}{2}y$ and therefore we have that:



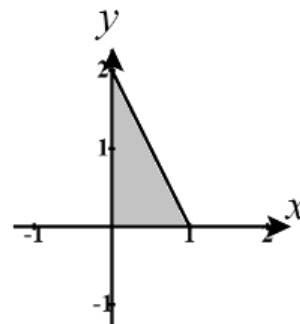
$$\begin{aligned} \iint_D (\mathbf{F} \cdot \nabla G) dA &= \iint_D \langle x^2, yx, zx \rangle \cdot \left\langle 3, \frac{3}{2}, 1 \right\rangle dA = \iint_D \left(3x^2 + \frac{3}{2}yx + zx \right) dA = \iint_D \left(3x^2 + \frac{3}{2}yx + \left(3 - 3x - \frac{3}{2}y \right)x \right) dA \\ &= \iint_D \left(3x^2 + \frac{3}{2}yx + 3x - 3x^2 - \frac{3}{2}xy \right) dA \\ &= \iint_D 3x dA \end{aligned}$$

The region D is the projection of the solid into the xy -plane as shown in the picture to the right.

$$D = \{(x, y) | 0 \leq y \leq 2 - 2x \text{ \& } 0 \leq x \leq 1\}$$

Therefore, we have that:

$$\begin{aligned} \iint_D (\mathbf{F} \cdot \nabla G) dA &= \iint_D 3x dA = \int_0^1 \int_0^{2-2x} 3x dy dx \\ &= \int_0^1 3xy \Big|_0^{2-2x} dx = \int_0^1 3x(2-2x) dx = \int_0^1 (6x - 6x^2) dx \\ &= 3x^2 - 2x^3 \Big|_0^1 \\ &= 1 \end{aligned}$$



3. Let σ be the portion of the paraboloid $z = x^2 + y^2$ below the plane $z = 4$ oriented by inward normals and $\mathbf{F}(x, y, z) = \mathbf{i} + y\mathbf{j} + \mathbf{k}$. Compute the flux of $\mathbf{F}$ across σ in the direction of orientation.

Since σ represents the graph of a function of the form $z = f(x, y)$ we do not have to find a parameterization for σ and instead we tackle the problem as follows.

We note the function $G(x, y, z) = z - x^2 - y^2$ and therefore we have that:

$$\begin{aligned}\iint_D (\mathbf{F} \cdot \nabla G) dA &= \iint_D \langle 1, y, 1 \rangle \cdot \langle -2x, -2y, 1 \rangle dA \\ &= \iint_D (-2x - 2y^2 + 1) dA\end{aligned}$$

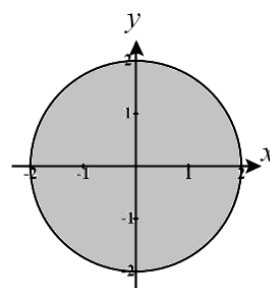
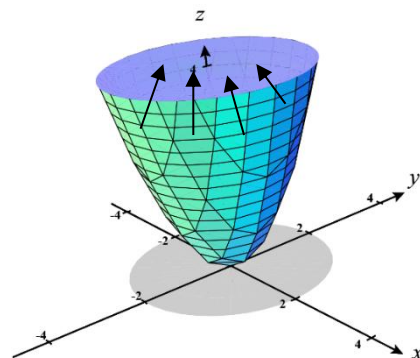
The region D is the projection of the solid into the xy -plane as shown in the picture to the right.

$$D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq 2\}$$

Therefore, we have that:

$$\begin{aligned}\iint_D (\mathbf{F} \cdot \nabla G) dA &= \iint_D (-2x - 2y^2 + 1) dA \\ &= \int_0^{2\pi} \int_0^2 (-2r \cos(\theta) - 2r^2 \sin^2(\theta) + 1) r dr d\theta \\ &= \int_0^{2\pi} \int_0^2 (-2r^2 \cos(\theta) - 2r^3 \sin^2(\theta) + r) dr d\theta \\ &= \int_0^{2\pi} \left(-\frac{2}{3} r^3 \cos(\theta) - \frac{1}{2} r^4 \sin^2(\theta) + \frac{1}{2} r^2 \right) \Big|_0^2 d\theta \\ &= \int_0^{2\pi} \left(-\frac{16}{3} \cos(\theta) - 8 \sin^2(\theta) + 2 \right) d\theta \\ &= \int_0^{2\pi} \left(-\frac{16}{3} \cos(\theta) - 4[1 - \cos(2\theta)] + 2 \right) d\theta \\ &= \left(-\frac{16}{3} \sin(\theta) - 4\theta + 2 \sin(2\theta) + 2\theta \right) \Big|_0^{2\pi} = \left(-\frac{16}{3} \sin(\theta) - 2\theta + 2 \sin(2\theta) \right) \Big|_0^{2\pi} \\ &= (0 - 4\pi + 0) - (0 - 0 + 0) \\ &= -4\pi\end{aligned}$$

Note: This result the opposite of the result we obtained on Example 5 in section 5.6. This is not surprising since we have simply changed the orientation of the normal vectors.



4. Let σ be the portion of the sphere $x^2 + y^2 + z^2 = 1$ that is inside the cone $z = \sqrt{\frac{1}{3}x^2 + \frac{1}{3}y^2}$ and $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Find the flux of $\mathbf{F}$ across σ with respect to upward normals.

We see that it will be difficult to write this surface as a function $z = f(x, y)$ and determine appropriate bounds on x and y . Therefore, we must approach this question by viewing the surface parametrically.

We create a parameterization for σ using our knowledge of spherical coordinates:

$$\mathbf{r}(u, v) = \langle \cos(u)\sin(v), \sin(u)\sin(v), \cos(v) \rangle \text{ where}$$

$0 \leq u \leq 2\pi$ & $0 \leq v \leq \frac{\pi}{3}$. (work may need to be done to demonstrate bounds on ϕ as done in section 4.6.2).

We calculate that:

$$\mathbf{r}_u = \langle -\sin(u)\sin(v), \cos(u)\sin(v), 0 \rangle$$

$$\mathbf{r}_v = \langle \cos(u)\cos(v), \sin(u)\cos(v), -\sin(v) \rangle$$

Therefore,

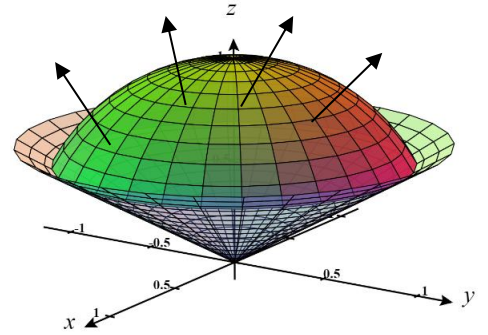
$$\begin{aligned} \mathbf{r}_u \times \mathbf{r}_v &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\sin(u)\sin(v) & \cos(u)\sin(v) & 0 \\ \cos(u)\cos(v) & \sin(u)\cos(v) & -\sin(v) \end{vmatrix} \\ &= \langle -\cos(u)\sin^2(v), -\sin(u)\sin^2(v), -\sin^2(u)\cos(v)\sin(v) - \cos^2(u)\cos(v)\sin(v) \rangle \\ &= \langle -\cos(u)\sin^2(v), -\sin(u)\sin^2(v), -\cos(v)\sin(v) \rangle \end{aligned}$$

Since we are working with upward facing normal and the z-component here is negative, we must use

$$-(\mathbf{r}_u \times \mathbf{r}_v) = \langle \cos(u)\sin^2(v), \sin(u)\sin^2(v), \cos(v)\sin(v) \rangle$$

Now, we have that,

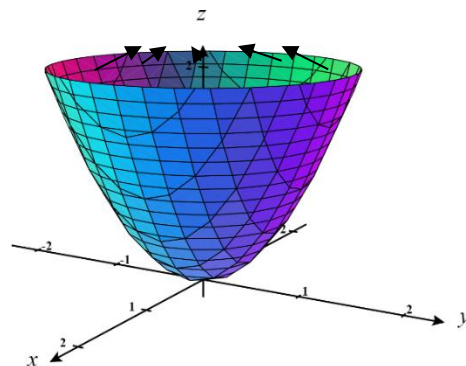
$$\begin{aligned} \iint_{\sigma} \mathbf{F} \cdot \mathbf{n} \, dS &= \iint_D \mathbf{F} \cdot -(\mathbf{r}_u \times \mathbf{r}_v) \, dA \\ &= \int_0^{\pi/3} \int_0^{2\pi} \langle \cos(u)\sin(v), \sin(u)\sin(v), \cos(v) \rangle \cdot \langle \cos(u)\sin^2(v), \sin(u)\sin^2(v), \cos(v)\sin(v) \rangle \, du \, dv \\ &= \int_0^{\pi/3} \int_0^{2\pi} [+\cos^2(u)\sin^3(v) + \sin^2(u)\sin^3(v) + \cos^2(v)\sin(v)] \, du \, dv \\ &= \int_0^{\pi/3} \int_0^{2\pi} [+\sin^3(v) + \cos^2(v)\sin(v)] \, du \, dv = \int_0^{\pi/3} \int_0^{2\pi} [\sin(v)] \, du \, dv = \int_0^{\pi/3} u \sin(v) \Big|_0^{2\pi} \, dv \\ &= \int_0^{\pi/3} 2\pi \sin(v) \, dv = -2\pi \cos(v) \Big|_0^{\pi/3} \\ &= -2\pi \left(\frac{1}{2} \right) + 2\pi(1) \\ &= \pi \end{aligned}$$



5. Let $\mathbf{F}(x, y, z) = 2xy\mathbf{i} - y^2\mathbf{j} + 2z^3\mathbf{k}$. Find the inward flux of the surface that is bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 2$.

Since σ represents the graph of a function of the form $z = f(x, y)$ we do not have to find a parameterization for σ and instead we tackle the problem as follows.

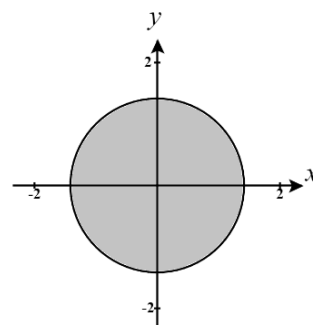
We note the function $G(x, y, z) = z - x^2 - y^2$. Since the normal vectors here are facing upward we have:



$$\begin{aligned}\iint_D (\mathbf{F} \cdot \nabla G) dA &= \iint_D \langle 2xy, -y^2, 2z^3 \rangle \cdot \langle -2x, -2y, 1 \rangle dA = \iint_D (-4x^2y + 2y^3 + 2z^3) dA \\ &= \iint_D (-4x^2y + 2y^3 + 2(x^2 + y^2)^3) dA\end{aligned}$$

The region D is the projection of the solid into the xy -plane as shown in the picture to the right.

$$D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq \sqrt{2}\}$$



Therefore, we have that:

$$\begin{aligned}\iint_D (\mathbf{F} \cdot \nabla G) dA &= \iint_D (-4x^2y + 2y^3 + 2(x^2 + y^2)^3) dA \\ &= \int_0^{2\pi} \int_0^{\sqrt{2}} (-4r^3 \cos^2(\theta) \sin(\theta) + 2r^3 \sin^3(\theta) + 2r^6) r dr d\theta \\ &= \int_0^{2\pi} \int_0^{\sqrt{2}} (-4r^4 \cos^2(\theta) \sin(\theta) + 2r^4 \sin^3(\theta) + 2r^7) dr d\theta \\ &= \int_0^{2\pi} \left(-\frac{4}{5} r^5 \cos^2(\theta) \sin(\theta) + \frac{2}{5} r^5 \sin^3(\theta) + \frac{1}{4} r^8 \right) \Big|_0^{\sqrt{2}} d\theta \\ &= \int_0^{2\pi} \left(-\frac{16\sqrt{2}}{5} \cos^2(\theta) \sin(\theta) + \frac{8\sqrt{2}}{5} \sin^3(\theta) + 4 \right) d\theta \\ &= \left[-\frac{16\sqrt{2}}{5} \int_0^{2\pi} \cos^2(\theta) \sin(\theta) d\theta + \frac{8\sqrt{2}}{5} \int_0^{2\pi} \sin^3(\theta) d\theta + \int_0^{2\pi} 4 d\theta \right]\end{aligned}$$

For the first integral we let $u = \cos(\theta)$ which implies $-du = \sin(\theta) d\theta$ and the bounds become $u(0) = 1$ & $u(2\pi) = 1$.

For the second integral we rewrite $\sin^3(\theta) = \sin^2(\theta)\sin(\theta) = [1 - \cos^2(\theta)]\sin(\theta)$ and then let $u = \cos(\theta)$ which implies $-du = \sin(\theta)d\theta$ and the bounds become $u(0) = 1$ & $u(2\pi) = 1$. Therefore, we have that,

$$\begin{aligned}
 & \left[-\frac{16\sqrt{2}}{5} \int_0^{2\pi} \cos^2(\theta) \sin(\theta) d\theta + \frac{8\sqrt{2}}{5} \int_0^{2\pi} \sin^3(\theta) d\theta + \int_0^{2\pi} 4d\theta \right] \\
 &= \left[-\frac{16\sqrt{2}}{5} \int_1^1 -u^2 du + \frac{8\sqrt{2}}{5} \int_1^1 -[1-u^2] du + \int_0^{2\pi} 4d\theta \right] \\
 &= -\frac{16\sqrt{2}}{5}(0) + \frac{8\sqrt{2}}{5}(0) + 8\pi \\
 &= 8\pi
 \end{aligned}$$